



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO NKK FUELS ADHOC

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 158.22 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.563933, 78.178055)



Welcome to the Journey Risk Management Study

To provide a comprehensive route safety analysis from Sangagiri, Tamil Nadu to Kundarapalli, Tamil Nadu, I'll address each of the factors listed:

1. Overview of the Route Map:

- The route spans approximately 158.22 kilometers through various roads including national highways. It starts from Sangagiri, primarily taking NH 544 before connecting to NH 44 and then

NH 48 towards Kundarapalli. The terrain is generally flat, allowing for smoother travel with some sections passing through hilly areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate with hot summers, a significant monsoon season (June to September), and mild winters. There's potential for heavy rain, flooding, and reduced visibility during the monsoon, which can lead to slippery roads and slower travel times.

3. Analysis of Traffic Patterns:

- Traffic congestion is common in larger towns and cities along the route during peak hours, typically between 8-10 AM and 5-8 PM. Watch for bottlenecks near urban areas including Salem and Krishnagiri, where commercial activity is high.

4. Assessment of Road Quality and Infrastructure:

- The primary highways (NH 544 and NH 48) are generally well-maintained, with sufficient signage and lighting. However, some local roads may have potholes or uneven surfaces, particularly in less developed areas.

5. Suggestions for Alternative Routes for Emergencies:

- If NH 48 or NH 44 is inaccessible, consider diverting through smaller state highways that run parallel to the main highways. However, these might be narrower and less maintained, so exercise caution and verify accessibility.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Regulations in Tamil Nadu require specific permits for transporting hazardous materials. Speed limits might be lower for trucks carrying such goods, especially through urban or environmentally sensitive areas. Adherence to the Indian Motor Vehicles Act is necessary.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- There have been reported incidents of tanker truck accidents in the region, often related to improper handling or poor road conditions. Such incidents tend to cluster around major junctions and densely populated areas.

8. Environmental Considerations and Sensitive Areas:

- The route passes through several agricultural zones and near forested areas, particularly in the hilly segments. It is essential to minimize emissions and avoid any material spillage to protect local biodiversity.

9. Analysis of Communication Coverage:

- While major highways have good cellular and internet coverage, there might be dead zones in more remote or hilly areas. GPS devices should be frequently updated and satellite phones considered as a backup.

10. Estimated Emergency Response Times:

- In urban areas like Salem, expect emergency services to respond within 30 minutes. In more rural sections, response times could range from 1 to 2 hours due to distance and road accessibility.

11. Overall Summary of Risk Assessment:

- The route is generally safe but requires caution due to potential congestion and weather-related impacts. Proper adherence to traffic laws, enhanced vigilance during peak hours and adverse weather, and preparedness for emergencies are vital for a safe journey.

Trucking companies are advised to train drivers on emergency procedures, continuously monitor weather conditions, and coordinate with local authorities to stay updated on road conditions and regulations specific to hazardous material transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	High	11.47403, 77.86841	15 KM/Hr
6	Turn	High	11.49149, 77.89601	15 KM/Hr
7	Turn	High	11.49224, 77.89601	15 KM/Hr
8	Turn	Medium	11.53501, 77.95707	30 KM/Hr
9	Turn	Medium	11.53601, 77.95856	30 KM/Hr
10	Turn	Medium	11.62403, 78.10893	30 KM/Hr
11	Turn	High	11.64180, 78.11808	15 KM/Hr
12	Turn	High	11.64325, 78.11993	15 KM/Hr
13	Turn	Medium	11.69519, 78.10775	30 KM/Hr
14	Turn	Medium	11.73958, 78.05105	30 KM/Hr
15	Turn	Medium	11.75835, 78.05016	30 KM/Hr
16	Turn	Medium	11.86847, 78.08367	30 KM/Hr
17	Turn	Medium	11.87942, 78.08197	30 KM/Hr
18	Turn	Medium	11.92585, 78.06507	30 KM/Hr
19	Turn	Medium	11.93704, 78.05402	30 KM/Hr
20	Turn	High	11.95818, 78.04978	15 KM/Hr
21	Turn	Medium	11.95988, 78.04948	30 KM/Hr
22	Turn	Medium	11.96304, 78.05136	30 KM/Hr
23	Turn	High	11.96399, 78.05135	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
24	Turn	High	11.96580, 78.04923	15 KM/Hr
25	Turn	Medium	11.96675, 78.04939	30 KM/Hr
26	Turn	Medium	11.96791, 78.05070	30 KM/Hr
27	Turn	Medium	12.19881, 78.18157	30 KM/Hr
28	Turn	Medium	12.22459, 78.19136	30 KM/Hr
29	Turn	Medium	12.25060, 78.19977	30 KM/Hr
30	Turn	Medium	12.25329, 78.20039	30 KM/Hr
31	Turn	Medium	12.29110, 78.19909	30 KM/Hr
32	Turn	Medium	12.39904, 78.21764	30 KM/Hr
33	Turn	Medium	12.40156, 78.21845	30 KM/Hr
34	Turn	Medium	12.40589, 78.22275	30 KM/Hr
35	Turn	Medium	12.50553, 78.22318	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
14	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
28	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
33	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
34	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
35	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
37	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
42	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
44	clinic	Shri Narayana Bone & Joint Clinic	12.523617, 78.210613	30 km/h	Medium
45	clinic	AZ Family Clinic	12.5234942, 78.2127422	30 km/h	Medium
47	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
48	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
49	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
30	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
36	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
38	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
39	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
40	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
41	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
43	school	Krishland Higher Secondary School	12.5181566, 78.2178556	30 km/h	Medium
46	school	Vivekananda School	12.5231937, 78.2121429	30 km/h	Medium
50	school	Sri Saraswathi Vidyalaya	12.5620799, 78.183049	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47403, 77.86841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73958, 78.05105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22459, 78.19136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25060, 78.19977



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25329, 78.20039



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29110, 78.19909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39904, 78.21764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40156, 78.21845



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40589, 78.22275



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50553, 78.22318